Deep Spatial-Temporal Alignment for Robotics Inspection Platforms

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Abstract

Spatial localization and temporal alignment are fundamental challenges in automated robotic inspection pipelines. In this paper, we propose a robust differential odometry framework paired with elastic Dynamic Time Warping (DTW) for spatial trajectory synchronization. Extensive experimental results demonstrate sub-centimeter alignment accuracy across complex constrained environments.

1 Introduction

Robotic inspection in confined industrial environments, such as HVAC ducts and pipelines, demands precise tracking under severe odometry slip and GPS-denied constraints [1]. Traditional time-based video analysis fails due to non-uniform vehicle velocity profiles.

2 Methodology

2.1 Kinematic State Estimation

Given left and right wheel encoder displacements Δs_L and Δs_R, the discrete pose increment is integrated over circular swept arcs:

$$\Delta\theta = \frac{\Delta s_R - \Delta s_L}{W}, \quad \Delta s = \frac{\Delta s_L + \Delta s_R}{2} \tag{1}$$

2.2 Trajectory Warping via DTW

Let $A = \{a_1, \ldots, a_N\}$ and $B = \{b_1, \ldots, b_M\}$ denote two inspection runs. The spatial alignment cost matrix $D(i, j)$ is computed as:

$$D(i,j) = \mathrm{cost}(a_i, b_j) + \min\left(D(i-1,j), D(i,j-1), D(i-1,j-1)\right) \tag{2}$$

3 Experimental Evaluation

4 Conclusion

We presented a real-time localization and spatial-temporal alignment engine suitable for embedded edge devices. Future work will investigate vision-inertial graph SLAM integration.

Table 1: Trajectory Tracking and Alignment Error

Method	**RMSE (m)**	**Max Error (m)**
Linear Interpolation	0.142	0.310
Vanilla DTW	0.045	0.082
Ours (Constrained DTW)	**0.012**	**0.024**

References

[1] John Smith, Jane Doe, and Robert Johnson. Robotic inspection in gps-denied confined environments: A survey. *IEEE Transactions on Robotics and Automation*, 40:112–128, 2024.